

Stellenbosch University

Department of Industrial Engineering

Operations Research 415

Exercises in Preparation for Tutorial 5

Memo

Problem 3 on pages 953–955.

Consider the following transition probability matrix,

$$P = \begin{array}{c} \begin{array}{cc} & \begin{array}{cccc} NY & J & NE & Dissipate \end{array} \\ \begin{array}{c} NY \\ J \\ NE \end{array} & \left[\begin{array}{cccc} \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & \frac{2}{3} & \frac{1}{3} & 0 \end{array} \right] \end{array} \end{array}$$

Let NY be the steady state level (in tons) of pollution in New York, let NE be the steady state level of pollution in Newark, and let J be the steady state level of pollution in Jersey City. Equating flow in to flow out for each city yields the following three equations:

$$\begin{aligned} 1\,000 + \frac{1}{3}N_2 &= \frac{2}{3}N_1, \\ 100 + \frac{2}{3}N_3 &= \frac{2}{3}N_2, \text{ and} \\ 50 + \frac{1}{3}N_1 + \frac{1}{3}N_2 &= \frac{2}{3}N_3. \end{aligned}$$

Solving these equations, we obtain NY = 3 450, J = 3 900 and NE = 3 750.

Problem 4 on pages 953–955.

Let H = Money introduced (in billions of dollars) into the system at Vulcan each month,

N_1 = Steady state money level at Vulcan,

N_2 = Steady state money level at Romulanville, and

N_3 = Steady state money level at Klingonville.

The transition probabilities are captured in the matrix

$$P = \begin{matrix} & \begin{matrix} N_1 & N_2 & N_3 & Lost \end{matrix} \\ \begin{matrix} N_1 \\ N_2 \\ N_3 \end{matrix} & \begin{bmatrix} \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix} \end{matrix}.$$

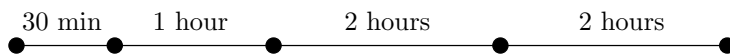
Therefore, we can derive the following set of equations:

$$\begin{aligned} H + \frac{1}{3}N_2 &= \frac{2}{3}N_1, \\ \frac{2}{3}N_3 &= \frac{2}{3}N_2, \text{ and} \\ \frac{1}{3}N_1 + \frac{1}{3}N_2 &= \frac{2}{3}N_3. \end{aligned}$$

We set $N_1 = N_2 = N_3 = 5$ and find the solution of $H = \frac{5}{3}$ satisfying these conditions. Therefore, by introducing \$1.67 million at Vulcan, one may achieve a steady-state level of \$5 billion in circulation at each planet.

Problem 2 on page 1063.

A typical segment of time is given in the figure below.



Thus, the chance of arriving during segments of various lengths are:

$\frac{30}{330} = 1/11$ for a 30 minute segment,

$\frac{60}{330} = 2/11$ for a 1 hour segment, and

$\frac{240}{330} = 8/11$ for a 2 hour segment.

The expected waiting time for the next bus may therefore be calculated as $\frac{1}{11}[0.5(30)] + \frac{2}{11}[0.5(60)] + \frac{8}{11}[0.5(120)] = 555/11 = 50.45$ minutes.

Note: Average time between buses = $\frac{1}{4}(30) + \frac{1}{4}(60) + \frac{1}{2}(120) = 82.5$ minutes.

Problem 3 on page 1063.

The average size of a third grade section is calculated as $\frac{1}{4}(20 + 25 + 35 + 40) = 30$ students. There is a $\frac{20}{120} = \frac{1}{6}$ chance that the selected student is from Section 1, a $\frac{25}{120} = \frac{5}{24}$ chance that the selected student is from Section 2, a $\frac{25}{120} = \frac{7}{24}$ chance that the selected student is from Section 3, and a $\frac{40}{120} = \frac{1}{3}$ chance that the selected student is from Section 4.

The average size of the selected student's class is therefore $\frac{1}{6}(20) + \frac{5}{24}(25) + \frac{7}{24}(35) + \frac{1}{3}(40) = \frac{770}{24} = 32.08$ students.

Problem 4 on page 1063.

(a) The number of buses arriving during the next two hours is Poisson distributed with a mean of 2.

$$P(4 \text{ buses during the next two hours}) = \frac{e^{-2}(2)^4}{4!} = 0.09.$$

(b) $P(\text{at least 2 buses}) = 1 - P(0 \text{ buses}) - P(1 \text{ bus}) = 1 - e^{-2} - 2e^{-2} = 0.59.$

(c) $P(0 \text{ buses}) = \frac{e^{-2}(2)^0}{0!} = e^{-2} = 0.14.$

(d) Interarrival time has density e^{-t} (in hours). Thus, we seek

$$\int_{1/2}^{3/2} e^{-t} dt = [-e^{-t}]_{1/2}^{3/2} = e^{-0.5} - e^{-1.5} = 0.38.$$

8. Problem 1 on page 1081.

We have an M/M/1 system with

$\lambda = 10$ customers/minute,

$\mu = 12$ customers/minute, and ρ

$$= 10/12 = 5/6.$$

(a) $1 - \pi_0 = 5/6.$

(b) $L_q = \frac{\rho^2}{1-\rho} = \frac{25/36}{1/6} = 25/6$ customers.

(c) $L = \frac{\rho}{1-\rho} = \frac{5/6}{1/6} = 5$ customers, and

$$W = \frac{L}{\lambda} = 5/10 = 1/2 \text{ minutes} = 1/120 \text{ hours}.$$

9. Problem 2 on page 1082.

For the slow copier service the cost/hour = \$4, $\lambda = 4$ customers/hour and $\mu = 6$ customers/hour. $W = \frac{1}{\mu - \lambda} = 1/2 \text{ hr.}$

Furthermore, the delay cost/hour = (4 customer/hr.) (1/2hr.) (\$15/cust./hr.) = \$30/hr.

Hence, the total cost/hour = \$34.

For the fast copier, service cost/hour = \$15/hr, $\lambda = 4$ customers/hour, and $\mu = 10$ customers/hr.

Therefore, $W = \frac{1}{\mu - \lambda} = 1/6 \text{ hr.}$

Furthermore, the delay cost/hour = (4 cust./hr.) (1/6 hr.) (\$15/cust./hr.) = \$10/hr.

Hence, the total cost/hour = \$25/hr.

Thus, the fast copier is the better option.

10. Problem 4 on page 1082.

We have an M/M/1/GD/ ∞/∞ system with $\lambda = 40$ customers per hour and $\mu = 60$ customers per

hour. Hence $\rho = \frac{\lambda}{\mu} = \frac{40}{60} = \frac{2}{3}.$

(a) $L_q = \frac{40^2}{(60-40)60} = 1.33$ customers.

(b) $W = \frac{1}{(60-40)} = 1/20 \text{ hour} = 3 \text{ minutes}.$

(c) $1 - \pi_0 - \pi_1 - \pi_2 - \pi_3 = 1 - 1/3 - 2/9 - 4/27 - 8/81 = 16/81.$

11. Problem 1 on page 1085.

We have an M/M/1/GD/3/ ∞ system with $\lambda = 3$ customers/hour and $\mu = 2$ customers/hour.

(a) We seek $\lambda(1 - \pi_3)$. Since

$$\pi_0 = \frac{1-3/2}{1-(3/2)^4} = \frac{16}{130},$$

$$\pi_3 = (3/2)^3 \left(\frac{16}{130} \right) = 0.415.$$

Thus, $\lambda(1 - \pi_3) = 1.75$ potential customers/hour enter the system.

(b) $1 - \pi_0 = \frac{114}{130}.$

12. **Problem 2 on page 1 085.**

We have an $M/M/1/GD/4/\infty$ system with $\mu = 15$ cars/hour and $\lambda = 40$ cars/hour.

(a) We seek $L_q = L - L_s$. Since $\rho = \frac{40}{15} = 2.67$, we may calculate

$$L = \frac{2.67[1-5(2.67)^4+4(2.67)^5]}{(1-2.67^5)(1-2.67)} = 3.44,$$

$$\pi_0 = \frac{1-2.67}{1-2.67^5} = 0.012, \text{ and}$$

$$L_s = 1 - \pi_0 = 0.99.$$

Therefore, $L_q = 3.44 - 0.99 = 2.45$.

(b) We seek $\mu(1 - \pi_0) = 15(1 - 0.012) = 14.82$ cars/hour (here $\pi_4 = (2.67)^4(0.012) = 0.61$).

(c) $W = \frac{L}{\lambda(1-\pi_4)} = \frac{3.44}{40(1-0.61)} = 0.22$ hours.

13. **Problem 6 on page 1 085.**

Each barber is an $M/M/1/GD/4/\infty$ system.

Barber 1 receives $\lambda = 10$ customers/hour and serves $\mu = 5$ customers/hour. Thus $\rho = \frac{10}{5} = 2$.

Hence, $\pi_0 = \frac{1-2}{1-2^5} = \frac{1}{31}$ and $\pi_4 = (2)^4(\frac{1}{31}) = \frac{16}{31}$.

Therefore, Barber 1's expected hourly revenue is $[1 - (\frac{16}{31})](10 \text{ customers/hour}) (\$11) = \$53.23$.

Barber 2 receives $\lambda = 10$ customers/hour and serves $\mu = 10$ customers/hour. Thus $\rho = 1$ and so all the π_j 's must equal $1/5$. Barber 2's expected hourly revenue is therefore $[1 - (1/5)](10 \text{ customers/hour})(\$5/\text{customer}) = \$40$.

Consequently, Barber 1 collects more revenue.

14. **Problem 1 on page 1 096.**

We have a $GI/G/\infty/GD/\infty/\infty$ system with

$\lambda = 100$ members/week and $W = 52$ weeks. Therefore, $L = \lambda W = 5200$ members.

15. **Problem 2 on page 1 096.**

Again we have a $GI/G/\infty/GD/\infty/\infty$ system with

$L =$ average number of doctoral students on campus and $\frac{1}{\mu} = W = 4$ years. Therefore, $L = 25(4) = 100$ students.

16 **Problem 1 on page 1 098.**

We have an $M/D/1/FCFS/\infty/\infty$ system. The Pollaczek-Khinchine formulae (on p. 1097) with $\sigma^2 = 0$, $\lambda = 20$ customers/hour and $\mu = 30$ customers/hour yield,

$$L_q = \frac{(20)^2(0)+(2/3)^2}{2(1-2/3)} = 2/3 \text{ customers and}$$

$$W_q = \frac{L_q}{\lambda} = \frac{(2/3)}{20} = 1/30 \text{ hour} = 2 \text{ minutes.}$$